A Tutor That Never Says “I Don’t Know”: Abstention Failure in a Retrieval-Augmented Question-Answering System and What It Means for Learners’ Trust

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Abstract

Retrieval-augmented generation (RAG) is often described as a way to make language-model answers more reliable, and that makes it an obvious design for AI tutors. For a learner, though, reliability is not only about accuracy. It also depends on whether the system lets you know when it has no good answer, since that is the signal a learner needs to calibrate trust. I built a RAG question-answering system from standard parts (a DPR retriever, a FAISS index, and Flan-T5-base) and tested it on a seeded 500-question slice of SQuAD 2.0, where 258 questions are labeled unanswerable. Retrieval raised exact match from 0.004 without retrieval to 0.062 with five retrieved passages. But the system never abstained. It gave an answer string for all 258 unanswerable questions, and its answers looked the same whether or not retrieval had found the right passage, which it did only for a minority of questions (Recall@10 = 0.310). I argue that this matters for education. A tutor that always answers takes away the cue a learner would use to tell supported output from unsupported output, and the usual aggregate metrics do not show the problem. I discuss what this means for how we evaluate educational AI and outline a study to test whether abstention helps learners rely on these systems appropriately.

Keywords: retrieval-augmented generation; abstention; unanswerable questions; trust calibration; appropriate reliance; AI in education

1. Introduction

Students now ask AI systems questions they cannot yet answer themselves. That is exactly when they are worst placed to judge whether the answer is right. Lee and See (2004) argued that automation often causes problems because people fail to rely on it appropriately, and that trust guides reliance when fully understanding the automation is impractical. The goal is appropriate reliance, not maximum reliance: using the system when it is right and not when it is wrong. Relying on automation when one should not is misuse, and failing to use it when it would help is disuse (Parasuraman & Riley, 1997). For a learner to rely appropriately, the system has to give some signal of when it is on solid ground.

RAG is a common answer to the reliability problem. Instead of drawing only on what a language model stores in its parameters, a RAG system retrieves passages from a document collection and conditions its answer on them (Lewis et al., 2020). For tutoring this is appealing, because the collection can be the course material. Retrieval alone, however, does not tell the system (or the learner) when the retrieved material does not contain an answer.

The dataset I used makes this concrete. SQuAD 2.0 was created because reading-comprehension systems "tend to make unreliable guesses on questions for which the correct answer is not stated in the context," and it requires systems to "determine when no answer is supported by the paragraph and abstain from answering" (Rajpurkar et al., 2018, abstract; the paper calls the dataset SQuADRUn, and it was released as SQuAD 2.0).

In this paper I report what happened when I built a RAG system from standard, publicly available components and ran it on SQuAD 2.0. It did better than a no-retrieval baseline. It also never once declined to answer. That second result is what I want to think through here. If a tutor answers every question the same way, whether or not it has any basis for the answer, what is left for a learner to calibrate trust against?

The contribution is small and specific. I document a clear case of abstention failure in a small, reproducible RAG pipeline, show how aggregate accuracy metrics hide it, connect it to research on how people rely on language models, and propose a study with learners. The system results are empirical. The claims about education are arguments from prior literature, not findings of this study, and I mark them that way.

2. Background

2.1 Abstention in question answering

A question-answering system can fail on an unanswerable question in two ways: it can say it cannot answer, or it can guess. SQuAD 2.0 added over 50,000 unanswerable questions, written adversarially by crowdworkers to look like answerable ones. A strong neural system that reached 86% F1 on the original SQuAD dropped to 66% F1 on SQuAD 2.0 (Rajpurkar et al., 2018).

Knowing when to abstain gets harder when inputs differ from the training data. Kamath et al. (2020) studied selective question answering under domain shift and found that abstention policies based only on a model's softmax probabilities do poorly, because models are overconfident on out-of-domain inputs. A separately trained calibrator worked better: at 80% accuracy it answered 56% of questions, compared with 48% when the model's own probabilities were used.

2.2 How people rely on language-model answers

Language models rarely signal doubt on their own. Zhou et al. (2024) found that deployed language models were reluctant to express uncertainty even when they were wrong, and that they were overconfident when prompted to state their confidence. In the same work, users relied heavily on model outputs whether or not certainty was marked.

People also overestimate how accurate these models are. Steyvers et al. (2025) found that people overestimated the accuracy of LLM answers when given default explanations, and that longer explanations raised confidence without improving accuracy. They separate a calibration gap (the difference between people's confidence in LLM answers and the model's own confidence) from a discrimination gap (the difference between how well people and the model tell correct from incorrect answers). In fact-checking, LLM explanations made users more efficient than a search engine did, but users overrelied on the model when its explanation was wrong (Si et al., 2024).

Communicating uncertainty can help, but the wording matters. In a pre-registered experiment, first-person expressions of uncertainty ("I'm not sure, but…") reduced agreement with an LLM-infused search engine and increased accuracy, while uncertainty phrased from a general perspective ("It's not clear, but…") had weaker effects that were not statistically significant (Kim et al., 2024). In an earlier decision-support study, showing confidence scores helped calibrate trust but did not by itself improve decisions (Zhang et al., 2020).

2.3 Learners as users

Learners are a special case, because the point is to learn, not just to get a correct answer. In a randomized trial with high school mathematics students, unrestricted access to GPT-4 improved performance during practice but hurt performance once access was removed. A version with safeguards, including teacher-designed hints in place of answers, largely removed that harm (Bastani et al., 2025). In lab sessions with 21 novice programmers, generative AI could compound the metacognitive difficulties of students who were already struggling, and those students overestimated their own performance (Prather et al., 2024). In a study of 432 undergraduates working on programming problems with an AI assistant, higher trust was associated with lower appropriate reliance in a non-linear relationship, and AI literacy moderated that relationship (Pitts et al., 2026).

Taken together, these studies suggest that learners need cues about when AI output is supported, and that we cannot assume those cues exist by default.

3. System

I built the system as a course project (Section 10), following the retrieve-then-generate design of RAG (Lewis et al., 2020): encode the question, retrieve the top-K passages, and generate an answer conditioned on them. Lewis et al. train the retriever and generator jointly and marginalize over passages. I did not. I concatenated the top-K passages into a single prompt, and I used every component off the shelf, without fine-tuning.

Retriever. Dense Passage Retrieval (DPR) uses separate BERT-based encoders for questions and passages and scores passages by the dot product of their embeddings (Karpukhin et al., 2020). I used the publicly released facebook/dpr-question_encoder-single-nq-base and facebook/dpr-ctx_encoder-single-nq-base checkpoints, which were trained on Natural Questions.

Index. Passage embeddings went into a FAISS IndexFlatIP index, which does exact maximum inner-product search (Johnson et al., 2021).

Generator. The generator was google/flan-t5-base, an instruction-finetuned T5 model released by Chung et al. (2024). I concatenated the top-K passages into one context and inserted it into the prompt "Answer the question. Context: {z} Question: {x} Answer:", truncated to 1,024 input tokens. Decoding used beam search with four beams and length normalization, with at most 32 new tokens. The prompt says nothing about what to do when the context does not contain an answer.

Corpus. The retrieval collection was about 20,000 unique passages that appear as contexts in SQuAD 2.0 (the validation split plus a seeded sample of the training split). I split them into windows of 120 tokens with a stride of 80, which gave about 29,000 chunks. Every evaluation question's source paragraph is in the collection, so a retrieval miss reflects the retriever, not missing coverage.

4. Evaluation

I evaluated on a fixed, seeded sample of 500 questions from the SQuAD 2.0 validation split: 242 answerable and 258 labeled unanswerable. I compared a closed-book condition (Flan-T5-base with no retrieved context) with RAG at K = 1, 3, 5, and 10 retrieved passages. I used three kinds of metrics:

- Exact match (EM) and token-level F1 against gold answers, after standard normalization.
- Recall@K, the share of questions whose gold passage appears among the top K retrieved passages.
- Answerability detection at K = 5. I classified each output as an answer or an abstention and computed precision, recall, and F1 for the answerable class.

All runs used a single Google Colab T4 GPU. I report one set of results per condition, with no variation across runs.

5. Results

5.1 Retrieval improved accuracy, from a very low base

Table 1 and Figure 1 summarize the results. Closed-book EM was 0.004, which is 2 of 500 questions. With retrieval, EM rose to 0.034 at K = 1, 0.056 at K = 3, and 0.062 at K = 5 and K = 10. F1 rose from 0.027 to 0.091 at K = 10.

In relative terms the gain at K = 5 is 15.5-fold, but I would not lean on that number, since the baseline is two correct answers. The absolute gain is 5.8 percentage points of EM, and the 95% intervals for closed-book and K = 5 EM do not overlap (Table 1).

Table 1. Answer quality and retrieval by condition (n = 500).

Condition	EM	95% CI for EM	F1	Recall@K
Closed-book	0.004	0.001–0.014	0.027	n/a
RAG, K = 1	0.034	0.021–0.054	0.058	0.116
RAG, K = 3	0.056	0.039–0.080	0.081	0.194
RAG, K = 5	0.062	0.044–0.087	0.089	0.215
RAG, K = 10	0.062	0.044–0.087	0.091	0.310

Note. Confidence intervals are Wilson score intervals, which I computed from the reported proportions (n = 500).

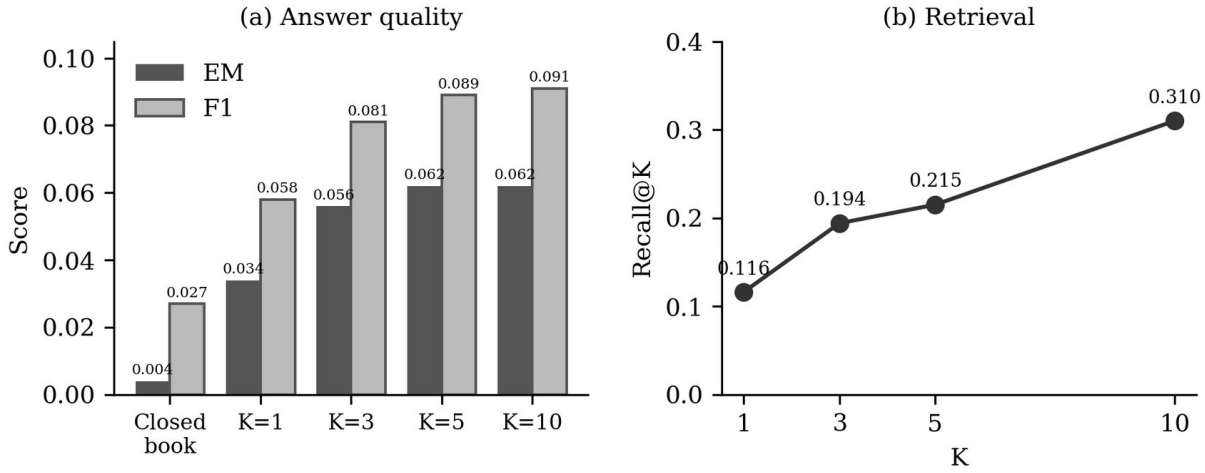


Figure 1. (a) Exact match and F1 for the closed-book baseline and RAG with K retrieved passages. (b) Recall@K of the DPR retriever.

5.2 Retrieval usually missed the source passage

Recall@1 was 0.116 and Recall@10 was 0.310. Even with ten passages, the source paragraph was missing from the context for most questions.

These numbers are not directly comparable to the retrieval accuracy Karpukhin et al. (2020) report. Their measure counts a hit when any retrieved passage contains the answer string, over a 21-million-passage Wikipedia collection. Their results do show that SQuAD is hard for DPR, though. Even a DPR trained on SQuAD reached 63.2% top-20 accuracy on SQuAD, below the 68.8% of the lexical baseline BM25. The authors give two reasons: SQuAD annotators wrote questions after seeing the passage, which creates high lexical overlap that favors BM25, and the data come from only 500+ Wikipedia articles (Karpukhin et al., 2020). My retriever was trained on Natural Questions and never saw SQuAD, so low recall is not a surprise.

5.3 The system never abstained

At K = 5, not a single output was an abstention (Table 2). All 258 unanswerable questions got an answer string. Precision for the answerable class was 242/500 = 0.484, recall was 1.000, and F1 was 0.652.

An F1 of 0.652 looks moderate. It comes entirely from answering everything, and a system with no ability at all to detect unanswerable questions gets the same score.

Table 2. Answerability at K = 5.

Gold label	Predicted: abstain	Predicted: answer
Answerable (n = 242)	0	242
Unanswerable (n = 258)	0	258

5.4 Wrong answers looked like right ones

Table 3 shows three examples from the evaluation slice. In the first, retrieval led to a correct extractive answer that the closed-book model missed. In the second, both conditions were wrong. In the third, the

question is labeled unanswerable, and both conditions still gave a specific number. Nothing in the form of the output separates the first case from the third.

Table 3. Example outputs.

Question	Gold answer	Closed-book	RAG (K = 5)
Cascade type when complement proteins bind microbes?	catalytic cascade	cytochrome c cascade (wrong)	catalytic (correct)
Main defense mechanism of bacteria?	restriction modification system	prokaryote (wrong)	antimicrobial peptide (wrong)
How many beds at Maria Memorial?	Unanswerable	84	78

Note. Questions are shortened as they appeared in my project materials.

6. Discussion: What This Means for Learners

Everything above is a measurement of one system. This section is interpretation. It links those measurements to prior research on reliance, and none of it was tested with learners.

6.1 An answer to every question removes a cue

Imagine putting a tutor built like this in front of students. At K = 5, the source paragraph was among the retrieved passages about one time in five (Recall@5 = 0.215), and the answer was exactly right for about one question in sixteen (EM = 0.062). Yet every question got an answer, and the system returns only the generated text, with no sign of confidence. For the 258 questions with no supported answer, every output was unsupported and none was flagged.

A learner using this system has nothing in its behavior to tell them when to trust it, and prior work suggests learners are unlikely to make up for that on their own. Users tend to overestimate the accuracy of LLM answers that come with default explanations (Steyvers et al., 2025). They rely on outputs whether or not certainty is marked (Zhou et al., 2024), and they overrely when a model is convincingly wrong (Si et al., 2024). In one qualitative study, struggling students had a harder time than their faster peers setting aside incorrect AI suggestions, and they often overestimated their own performance (Prather et al., 2024). The answer to “Maria Memorial has how many beds?” (78) is the kind of confident, checkable-looking detail a student could easily copy into an assignment.

Retrieval might make this worse rather than better. A RAG tutor can truthfully say it grounds its answers in course materials. In my system, the source paragraph was missing from the context for most questions, even at K = 10 (Recall@10 = 0.310). The retrieval step ran every time. The grounding it implies mostly did not happen.

6.2 Aggregate metrics hid the failure

If I had reported only EM and F1, my headline would have been a 15.5-fold improvement from retrieval. I only saw the abstention failure once I scored answerable and unanswerable questions separately, and even then the detection F1 of 0.652 looked like partial success until I looked at the confusion matrix.

The trust literature makes a similar point about people. How well someone can tell correct from incorrect AI output is a different quantity from how confident they are on average (Steyvers et al., 2025). I think the same holds for systems. For an educational deployment, "how often is the tutor right?" is not enough. An evaluation should also ask how often the tutor answers when it should not, and report that number on its own.

6.3 Abstention as a teaching move

For a tutor, "I don't know" does not have to be a dead end. A reply like "I couldn't find this in your course materials" tells the learner something true about the system's limits and points them somewhere to look next. Research on uncertainty expression suggests the wording of such messages matters: first-person uncertainty reduced agreement with the system and increased accuracy, while uncertainty phrased from a general perspective had weaker effects that were not significant (Kim et al., 2024).

Whether this carries over to learners is an open question, and two concerns need testing. Abstention could lower learners' trust in correct answers along with incorrect ones. And in earlier work, confidence information alone did not improve decisions (Zhang et al., 2020). There is also a separate reason abstention might matter in education. Bastani et al. (2025) found that safeguards, especially teacher-designed hints in place of answers, largely removed the harm to learning seen with unrestricted access. A tutor that declines to answer when it has no basis might share some of that benefit. That is a hypothesis, not a finding.

7. A Proposed Study

To move from argument to evidence, I propose a between-subjects experiment with students. Participants would answer reading-comprehension questions with help from one of two versions of the tutor: the current system, or a version that abstains when retrieval confidence is low. The abstaining version could use a calibrator in the style of Kamath et al. (2020) or an explicit refusal instruction in the prompt, and I would first measure its abstention accuracy on held-out SQuAD 2.0 questions.

The question set would mix answerable questions, unanswerable questions, and answerable questions where the tutor is wrong. The main outcome would be appropriate reliance: agreeing with the tutor when it is correct and disagreeing when it is not. Secondary outcomes would be discrimination between correct and incorrect tutor outputs (Steyvers et al., 2025), self-reported trust, and performance on a later test without the tutor, following the design logic of Bastani et al. (2025). Because AI literacy moderates how trust turns into reliance (Pitts et al., 2026), I would also measure AI literacy and test it as a moderator.

8. Limitations

This was a course project run on one GPU. The results come from a single seeded 500-question sample, and I report no variation across runs or samples.

- **Small counts.** The closed-book baseline rests on two correct answers, so relative gains are unstable. Differences among the retrieval conditions (for example, EM 0.056 at K = 3 versus 0.062 at K = 5) fall within overlapping intervals.
- **Small, off-the-shelf components.** The retriever was never trained on SQuAD, and the generator is small. Stronger or fine-tuned systems would probably score higher, and some might abstain.

My finding is not that RAG systems cannot abstain. It is that a standard pipeline put together without an explicit abstention mechanism did not, and that the headline metrics did not show it.

- **Abstention mechanism.** Neither the prompt nor the decoding procedure supported abstention, so I cannot separate how much the prompt, the decoding settings, and the model each contributed to the failure.
- **Evaluation data.** The collection was built from SQuAD contexts, not course materials, and SQuAD questions are not student questions.
- **No learners.** No learners took part. Every claim about learners in Section 6 is an inference from prior studies.

9. Conclusion

A RAG question-answering system built from standard components did better than a no-retrieval baseline. It also answered all 258 questions it had no basis to answer, in exactly the same form as its correct answers. For education, I think the second result matters more than the first. A tutor that never says “I don’t know” gives learners nothing to calibrate against, and the metrics people usually report would not catch it. I suggest that educational AI be evaluated on whether it abstains when it should, and that abstention be designed into tutors from the start rather than added later as a technical fix.

10. Declarations

Origin of this work. The system and all results in Sections 3 to 5 come from my final project for CSCI E-222, Foundations of Large Language Models, at Harvard University (Spring 2026). This paper reframes that project around a question about education. The literature review, the confidence intervals, and Sections 6 and 7 are new.

Use of AI tools. Claude (Anthropic) helped draft the decision, revise the wording.

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